

Lab – ReactJS part 3

My Blog Website using React JS and Firebase

1. Project File structure

```

my-app/
├── .env.local
├── src/
│   ├── components/
│   │   ├── Navbar.jsx          <-- Header bar ("AdminPortal" brand &
toggle)
│   │   ├── Sidebar.jsx        <-- Offcanvas drawer menu (Admin links
only)
│   │   ├── Footer.jsx         <-- Site footer
│   │   └── DashboardComponents.jsx <-- Dynamic StatCards & Recent Posts
Table
│   │   ├── PostForm.jsx       <-- Rich Text HTML Form (URL-based images)
│   │   └── PostList.jsx       <-- Post table with "+ Create New Post"
button
│   ├── lib/
│   │   └── firebaseClient.js  <-- Firebase Auth & Firestore client
│   ├── pages/
│   │   ├── HomePage.jsx       <-- Public blog card list (/)
│   │   ├── PostDetailPage.jsx <-- Full post reader view (/post/:id)
│   │   ├── AdminPortalPage.jsx <-- AdminPortal main dashboard (/admin)
│   │   └── PostsManager.jsx    <-- Toggles between Table List & Form
(/admin/posts)
│   ├── LoginPage.jsx          <-- Login page
│   ├── RegisterPage.jsx       <-- Registration page
│   ├── App.jsx                <-- Routing & Global Auth State
│   ├── main.jsx               <-- React entry point with BrowserRouter
│   └── index.css

```

2. Terminal Setup & Installation

Run this command in your project terminal to install the necessary packages

```
npm install react-router-dom firebase react-quill-new
```

3. Environment Variables (.env.local)

```

VITE_FIREBASE_API_KEY="your_api_key_here"
VITE_FIREBASE_AUTH_DOMAIN="your_project_id.firebaseio.com"
VITE_FIREBASE_PROJECT_ID="your_project_id"

```

```
VITE_FIREBASE_STORAGE_BUCKET="your_project_id.firebasestorage.app"
VITE_FIREBASE_MESSAGING_SENDER_ID="your_messaging_sender_id"
VITE_FIREBASE_APP_ID="your_app_id"
```

4. Firestore Production Security Rules

In your **Firebase Console** → **Firestore Database** → **Rules**, publish these security rules: `rules_version = '2';`

```
service cloud.firestore {
  match /databases/{database}/documents {
    match /posts/{postId} {
      allow read: if true;
      allow write: if request.auth != null;
    }
  }
}
```

5. Complete Source Code

src/lib/firebaseClient.js

```
// src/lib/firebaseClient.js
import { initializeApp } from "firebase/app";
import { getAuth } from "firebase/auth";
import { getFirestore } from "firebase/firestore";

const firebaseConfig = {
  apiKey: import.meta.env.VITE_FIREBASE_API_KEY,
  authDomain: import.meta.env.VITE_FIREBASE_AUTH_DOMAIN,
  projectId: import.meta.env.VITE_FIREBASE_PROJECT_ID,
  storageBucket: import.meta.env.VITE_FIREBASE_STORAGE_BUCKET,
  messagingSenderId: import.meta.env.VITE_FIREBASE_MESSAGING_SENDER_ID,
  appId: import.meta.env.VITE_FIREBASE_APP_ID
};

const app = initializeApp(firebaseConfig);
export const auth = getAuth(app);
export const db = getFirestore(app);
```

src/main.jsx

```
import React from "react";
import ReactDOM from "react-dom/client";
import { BrowserRouter } from "react-router-dom";
import App from "./App.jsx";
```

```
import "./index.css";

ReactDOM.createRoot(document.getElementById("root")).render(
  <React.StrictMode>
    <BrowserRouter>
      <App />
    </BrowserRouter>
  </React.StrictMode>
);
```

src/App.jsx

```
import { useEffect, useState } from "react";
import { Routes, Route, Navigate } from "react-router-dom";
import { auth } from "../lib/firebaseClient";
import { onAuthStateChanged, signOut } from "firebase/auth";
import Navbar from "../components/Navbar";
import Sidebar from "../components/Sidebar";
import Footer from "../components/Footer";
import HomePage from "../pages/HomePage";
import PostDetailPage from "../pages/PostDetailPage";
import AdminPortalPage from "../pages/AdminPortalPage";
import PostsManager from "../pages/PostsManager";
import LoginPage from "../pages/LoginPage";
import RegisterPage from "../pages/RegisterPage";

export default function App() {
  const [user, setUser] = useState(null);
  const [checkingSession, setCheckingSession] = useState(true);
  const [isSidebarOpen, setIsSidebarOpen] = useState(true);

  useEffect(() => {
    const unsubscribe = onAuthStateChanged(auth, (currentUser) => {
      setUser(currentUser || null);
      setCheckingSession(false);
      setIsSidebarOpen(!!currentUser);
    });
    return () => unsubscribe();
  }, []);

  async function handleLogout() {
    try {
      await signOut(auth);
      setIsSidebarOpen(false);
    } catch (err) {
      console.error("Error signing out:", err);
    }
  }

  if (checkingSession) {
    return (
```

```

    <div className="min-h-screen flex items-center justify-center bg-
gray-50">
      <p className="text-gray-600 animate-pulse font-medium">Checking
session...</p>
    </div>
  );
}

return (
  <div className="flex flex-col min-h-screen bg-gray-50 text-gray-800">
    <Navbar
      user={user}
      onLogout={handleLogout}
      isSidebarOpen={isSidebarOpen}
      setIsSidebarOpen={setIsSidebarOpen}
    />

    <div className="flex flex-1 pt-16 relative">
      {user && (
        <Sidebar
          user={user}
          isOpen={isSidebarOpen}
          setIsOpen={setIsSidebarOpen}
        />
      )}

      <main className={`flex-1 p-4 md:p-6 transition-all duration-300
${user && isSidebarOpen ? 'md:pl-64' : 'md:pl-0'}}`>
        <Routes>
          {/* Public Routes */}
          <Route path="/" element={<HomePage />} />
          <Route path="/post/:id" element={<PostDetailPage />} />

          {/* Admin Portal Protected Routes */}
          <Route
            path="/admin"
            element={user ? <AdminPortalPage user={user} /> : <Navigate
to="/login" replace />}
          />
          <Route
            path="/admin/posts"
            element={user ? <PostsManager user={user} /> : <Navigate
to="/login" replace />}
          />

          {/* Auth Routes */}
          <Route
            path="/login"
            element={!user ? <LoginPage /> : <Navigate to="/admin"
replace />}
          />
          <Route
            path="/register"
            element={!user ? <RegisterPage /> : <Navigate to="/admin"

```

```

    replace />}
        />
    </Routes>
  </main>
</div>

  <div className={`transition-all duration-300 ${user && isSidebarOpen
? 'md:pl-64' : 'md:pl-0'}}`>
    <Footer />
  </div>
</div>
);
}

```

src/components/Navbar.jsx

```

import { Link } from "react-router-dom";

export default function Navbar({ user, onLogout, isSidebarOpen,
setIsSidebarOpen }) {
  return (
    <header className="bg-white border-b border-gray-200 fixed top-0 left-0
right-0 h-16 z-50">
      <nav className="h-full px-4 flex items-center justify-between">

        {/* Brand Logo links to /admin if logged in, or / if guest */}
        <div className="flex items-center gap-3">
          <Link
            to={user ? "/admin" : "/"}
            className="text-xl font-bold text-blue-600 tracking-tight"
          >
            {user ? "Admin Portal" : "My Blog"}
          </Link>

          {user && (
            <button
              onClick={() => setIsSidebarOpen(!isSidebarOpen)}
              className="p-2 rounded-lg hover:bg-gray-100 text-gray-600
transition-colors focus:outline-none"
              aria-label="Toggle Sidebar"
            >
              <svg className="w-6 h-6" fill="none" stroke="currentColor"
viewBox="0 0 24 24">
                {isSidebarOpen ? (
                  <path strokeLinecap="round" strokeLinejoin="round"
strokeWidth="2" d="M6 18L18 6M6 6L12 12" />
                ) : (
                  <path strokeLinecap="round" strokeLinejoin="round"
strokeWidth="2" d="M4 6h16M4 12h16M4 18h16" />
                )}
              </svg>
            </button>
          )}
        </div>
      </nav>
    </header>
  );
}

```

```

        </button>
      )}
    </div>

    { /* User Auth Controls */}
    <div className="flex items-center gap-4">
      { !user ? (
        <div className="flex items-center gap-2">
          <Link to="/register" className="px-3 py-1.5 rounded-md text-sm font-medium text-gray-600 hover:bg-gray-100 transition-colors">
            Register
          </Link>
          <Link to="/login" className="px-3 py-1.5 rounded-md text-sm font-semibold bg-blue-600 text-white hover:bg-blue-700 transition-colors shadow-sm">
            Login
          </Link>
        </div>
      ) : (
        <div className="flex items-center gap-4">
          <button
            onClick={onLogout}
            className="px-3 py-1.5 rounded-md text-sm font-semibold bg-red-600 text-white hover:bg-red-500 transition-colors shadow-sm"
          >
            Logout
          </button>
        </div>
      )}
    </div>
  </nav>
</header>
);
}

```

src/components/Sidebar.jsx

```

import { Link, useLocation } from "react-router-dom";

export default function Sidebar({ user, isOpen, setIsOpen }) {
  const location = useLocation();

  // Public site removed from sidebar
  const menuItems = [
    { name: "Admin Portal", path: "/admin" },
    { name: "Manage Posts", path: "/admin/posts" },
  ];

  const linkClass = "flex items-center px-4 py-3 text-sm font-medium text-gray-600 rounded-lg hover:bg-gray-100 hover:text-gray-900 transition-colors";

```

```

const activeClass = "flex items-center px-4 py-3 text-sm font-semibold
text-white bg-blue-600 rounded-lg shadow-sm transition-colors";

return (
  <>
    {isOpen && (
      <div
        className="fixed inset-0 bg-gray-900/30 backdrop-blur-sm z-40
md:hidden"
        onClick={() => setIsOpen(false)}
      />
    )}

    <aside
      className={`fixed top-16 bottom-0 left-0 w-64 bg-white border-r
border-gray-200 z-40 p-4 flex flex-col justify-between transition-transform
duration-300 ${
        isOpen ? 'translate-x-0' : '-translate-x-full'
      }}
    >
      <div className="space-y-6">
        <div className="px-4 text-xs font-semibold text-gray-400
uppercase tracking-wider">
          Admin Management
        </div>

        <nav className="space-y-1">
          {menuItems.map((item, index) => (
            <Link
              key={index}
              to={item.path}
              onClick={() => {
                if (window.innerWidth < 768) setIsOpen(false);
              }}
              className={location.pathname === item.path ? activeClass :
linkClass}
            >
              {item.name}
            </Link>
          ))}
        </nav>
      </div>

      <div className="border-t border-gray-200 pt-4 flex items-center
gap-3">
        <div className="w-9 h-9 rounded-full bg-blue-100 text-blue-600
flex items-center justify-center font-bold text-sm select-none">
          {user?.email?.charAt(0).toUpperCase()}
        </div>
        <div className="truncate max-w-[160px]">
          <p className="text-xs font-semibold text-gray-800 truncate">
            {user?.email}</p>
          <p className="text-[10px] text-gray-400 font-
medium">Administrator</p>

```

```

        </div>
      </div>
    </aside>
  </>
);
}

```

src/components/PostForm.jsx

```

import { useState, useEffect, useRef, useMemo } from "react";
import ReactQuill from "react-quill-new";
import "react-quill-new/dist/quill.snow.css";

export default function PostForm({ editingPost, onSubmit, onCancel }) {
  const [title, setTitle] = useState("");
  const [content, setContent] = useState("");
  const quillRef = useRef(null);

  useEffect(() => {
    if (editingPost) {
      setTitle(editingPost.title || "");
      setContent(editingPost.content || "");
    } else {
      setTitle("");
      setContent("");
    }
  }, [editingPost]);

  const handleImageUrlPrompt = () => {
    const url = window.prompt("Enter the image URL (e.g., https://example.com/image.jpg):");
    if (!url) return;

    const editor = quillRef.current.getEditor();
    const range = editor.getSelection(true);
    editor.insertEmbed(range.index, "image", url);
    editor.setSelection(range.index + 1);
  };

  const modules = useMemo(
    () => ({
      toolbar: {
        container: [
          [{ header: [1, 2, 3, false] }],
          ["bold", "italic", "underline", "strike", "blockquote"],
          [{ list: "ordered" }, { list: "bullet" }],
          ["link", "image"],
          ["clean"],
        ],
        handlers: {
          image: handleImageUrlPrompt,

```



```

    },
  },
  }),
  []
);

function handleSubmit(e) {
  e.preventDefault();
  if (!title.trim() || !content.trim()) return;

  onSubmit({ title, content });
  setTitle("");
  setContent("");
}

return (
  <div className="bg-white p-6 rounded-xl border border-gray-200 shadow-sm">
    <h2 className="text-lg font-semibold text-gray-800 mb-4">
      {editingPost ? "Edit Post" : "Create New Post"}
    </h2>

    <form onSubmit={handleSubmit} className="space-y-4">
      <div>
        <label className="block text-sm font-medium text-gray-700 mb-1">
          Post Title
        </label>
        <input
          type="text"
          value={title}
          onChange={(e) => setTitle(e.target.value)}
          placeholder="Enter post title..."
          required
          className="w-full rounded-lg border border-gray-300 px-4 py-2 text-sm focus:outline-none focus:ring-2 focus:ring-blue-600"
        />
      </div>

      <div>
        <label className="block text-sm font-medium text-gray-700 mb-1">
          Blog Content (Rich Text)
        </label>

        <div className="bg-white rounded-lg border border-gray-300">
          <ReactQuill
            ref={quillRef}
            theme="snow"
            value={content}
            onChange={setContent}
            modules={modules}
            placeholder="Write your blog post content here..."
            className="min-h-[200px]"
          />
        </div>
      </div>
    </form>
  </div>
);

```

```

    </div>

    <div className="flex items-center gap-3 pt-2">
      <button
        type="submit"
        className="px-4 py-2 bg-blue-600 text-white rounded-lg text-sm
font-semibold hover:bg-blue-700 transition-colors"
      >
        {editingPost ? "Update Post" : "Publish Post"}
      </button>

      {editingPost && (
        <button
          type="button"
          onClick={onCancel}
          className="px-4 py-2 bg-gray-200 text-gray-700 rounded-lg
text-sm font-semibold hover:bg-gray-300 transition-colors"
        >
          Cancel
        </button>
      )}
    </div>
  </form>
</div>
);
}

```

src/components/PostList.jsx

```

export default function PostList({ posts, loading, onCreateNew, onEdit,
onDelete }) {
  return (
    <div className="bg-white rounded-xl shadow-sm border border-gray-200
overflow-hidden space-y-4">
      {/* Table Header with "+ Create New Post" Button */}
      <div className="px-6 py-4 border-b border-gray-100 bg-gray-50/50 flex
justify-between items-center">
        <div>
          <h2 className="text-base font-semibold text-gray-800">Published
Posts</h2>
          <span className="text-xs text-gray-500">{posts.length} Total
Posts</span>
        </div>

        <button
          onClick={onCreateNew}
          className="px-4 py-2 bg-blue-600 text-white rounded-lg text-sm
font-semibold hover:bg-blue-700 transition-colors shadow-sm"
        >
          + Create New Post
        </button>
      </div>
    </div>
  );
}

```

```

    </div>

    {loading ? (
      <p className="p-6 text-sm text-gray-500 animate-pulse">Loading
posts...</p>
    ) : posts.length === 0 ? (
      <p className="p-6 text-sm text-gray-500">No blog posts found. Click
above to create your first post!</p>
    ) : (
      <div className="overflow-x-auto">
        <table className="min-w-full text-sm">
          <thead className="bg-gray-50 text-gray-600 border-b border-
gray-100">
            <tr>
              <th className="px-6 py-3 text-left font-
semibold">Title</th>
              <th className="px-6 py-3 text-left font-
semibold">Author</th>
              <th className="px-6 py-3 text-right font-
semibold">Actions</th>
            </tr>
          </thead>
          <tbody className="divide-y divide-gray-100 bg-white">
            {posts.map((post) => (
              <tr key={post.id} className="hover:bg-gray-50/50
transition-colors">
                <td className="px-6 py-4 font-medium text-gray-800">
{post.title}</td>
                <td className="px-6 py-4 text-gray-500">
{post.authorEmail}</td>
                <td className="px-6 py-4 text-right space-x-2">
                  <button
                    onClick={() => onEdit(post)}
                    className="px-3 py-1 bg-yellow-100 text-yellow-700
rounded-md text-xs font-semibold hover:bg-yellow-200 transition-colors"
                  >
                    Update
                  </button>
                  <button
                    onClick={() => onDelete(post.id)}
                    className="px-3 py-1 bg-red-100 text-red-700 rounded-
md text-xs font-semibold hover:bg-red-200 transition-colors"
                  >
                    Delete
                  </button>
                </td>
              </tr>
            )})}
          </tbody>
        </table>
      </div>
    )}
  </div>

```

```
);
}
```

src/components/DashboardComponents.jsx

```
import { Link } from "react-router-dom";

// Metric Display Card
export function StatCard({ title, value, color = "text-blue-600", isText =
false }) {
  return (
    <div className="bg-white rounded-xl border border-gray-200 shadow-sm p-
5">
      <p className="text-sm font-medium text-gray-500">{title}</p>
      <p className={`font-bold mt-2 ${isText ? "text-xl" : "text-3xl"}
${color}`}>
        {value}
      </p>
    </div>
  );
}

// Recent Posts Overview Table
export function RecentPostsTable({ posts, loading }) {
  return (
    <div className="bg-white rounded-xl shadow-sm border border-gray-200
overflow-hidden">
      <div className="px-6 py-4 border-b border-gray-100 flex items-center
justify-between bg-gray-50/50">
        <h2 className="text-base font-semibold text-gray-800">Recent Blog
Activity</h2>
        <span className="text-[10px] font-bold text-gray-500 uppercase
tracking-wider bg-gray-200/70 px-2 py-0.5 rounded">
          Live Data
        </span>
      </div>

      {loading ? (
        <p className="p-6 text-sm text-gray-500 animate-pulse">Loading
statistics...</p>
      ) : posts.length === 0 ? (
        <div className="p-6 text-center space-y-2">
          <p className="text-sm text-gray-500">No blog posts available in
the system yet.</p>
          <Link to="/admin/posts" className="text-xs text-blue-600 font-
semibold hover:underline">
            Create your first post →
          </Link>
        </div>
      ) : (
        <div className="overflow-x-auto">
```

```

        <table className="min-w-full text-sm">
          <thead className="bg-gray-50 text-gray-600 border-b border-
gray-100">
            <tr>
              <th className="px-6 py-3 text-left font-semibold">Post
Title</th>
              <th className="px-6 py-3 text-left font-
semibold">Author</th>
              <th className="px-6 py-3 text-left font-semibold">Published
Date</th>
              <th className="px-6 py-3 text-right font-
semibold">Action</th>
            </tr>
          </thead>
          <tbody className="divide-y divide-gray-100 bg-white">
            {posts.map((post) => (
              <tr key={post.id} className="hover:bg-gray-50/50
transition-colors">
                <td className="px-6 py-4 font-medium text-gray-800">
{post.title}</td>
                <td className="px-6 py-4 text-gray-500">
{post.authorEmail}</td>
                <td className="px-6 py-4 text-gray-500">
{post.createdAt?.toDate ?
post.createdAt.toDate().toLocaleDateString() : "Just now"}
</td>
                <td className="px-6 py-4 text-right">
                  <Link
                    to="/admin/posts"
                    className="px-3 py-1 bg-blue-50 text-blue-600
rounded-md text-xs font-semibold hover:bg-blue-100 transition-colors"
                  >
                    Manage
                  </Link>
                </td>
              </tr>
            )})}
          </tbody>
        </table>
      </div>
    )}
  </div>
);
}

```

src/components/Footer.jsx

```

export default function Footer() {
  return (
    <footer className="border-t border-gray-200 bg-white">
      <div className="max-w-6xl mx-auto px-4 py-6 flex items-center

```

```

justify-center">
  <p className="text-sm text-gray-500">
    © {new Date().getFullYear()} My Blog. All rights reserved.
  </p>
</div>
</footer>
);
}

```

src/pages/HomePage.jsx (Public Card Feed at /)

```

import { useState, useEffect } from "react";
import { Link } from "react-router-dom";
import { db } from "../lib/firebaseClient";
import { collection, getDocs, query, orderBy } from "firebase/firestore";

export default function HomePage() {
  const [posts, setPosts] = useState([]);
  const [loadingPosts, setLoadingPosts] = useState(true);

  useEffect(() => {
    async function fetchPublicPosts() {
      try {
        const q = query(collection(db, "posts"), orderBy("createdAt",
"desc"));
        const querySnapshot = await getDocs(q);
        const fetchedPosts = querySnapshot.docs.map((doc) => ({
          id: doc.id,
          ...doc.data(),
        }));
        setPosts(fetchedPosts);
      } catch (error) {
        console.error("Error fetching public posts:", error);
      } finally {
        setLoadingPosts(false);
      }
    }

    fetchPublicPosts();
  }, []);

  // Helper to extract first image URL or plain text snippet from raw HTML
  const extractThumbnail = (html) => {
    const doc = new DOMParser().parseFromString(html, "text/html");
    const img = doc.querySelector("img");
    return img ? img.src : null;
  };

  const extractSnippet = (html) => {
    const doc = new DOMParser().parseFromString(html, "text/html");
    const text = doc.body.textContent || "";
  }

```

```

    return text.length > 120 ? text.substring(0, 120) + "... " : text;
  };

  return (
    <div className="space-y-8 py-6 max-w-6xl mx-auto">
      <section className="text-center space-y-3">
        <h1 className="text-4xl font-extrabold text-gray-900 tracking-tight">
          Welcome to Our Blog
        </h1>
        <p className="text-lg text-gray-600">
          Discover latest updates, news, and insights.
        </p>
      </section>

      { /* Post Grid View */ }
      <div className="space-y-6">
        <h2 className="text-2xl font-bold text-gray-800 border-b pb-2">Latest Posts</h2>

        {loadingPosts ? (
          <p className="text-gray-500 animate-pulse">Loading blog posts...
        </p>
        ) : posts.length === 0 ? (
          <p className="text-gray-500">No blog posts published yet.</p>
        ) : (
          <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
            {posts.map((post) => {
              const thumbnail = extractThumbnail(post.content);
              const snippet = extractSnippet(post.content);

              return (
                <div key={post.id} className="bg-white rounded-xl border border-gray-200 shadow-sm overflow-hidden flex flex-col justify-between hover:shadow-md transition-shadow">
                  { /* Thumbnail Image */ }
                  <div className="h-48 bg-gray-100 w-full overflow-hidden flex items-center justify-center">
                    {thumbnail ? (
                      <img src={thumbnail} alt={post.title} className="w-full h-full object-cover" />
                    ) : (
                      <div className="text-gray-400 font-semibold text-sm">No Image Preview</div>
                    )}
                  </div>

                  { /* Card Content */ }
                  <div className="p-5 flex-1 flex flex-col justify-between space-y-4">
                    <div className="space-y-2">
                      <h3 className="text-xl font-bold text-gray-900 line-clamp-2">{post.title}</h3>

```

```

        <p className="text-sm text-gray-600 line-clamp-3">
{snippet}</p>
    </div>

    <div className="pt-2 border-t flex items-center
justify-between">
        <span className="text-xs text-gray-400">
            {post.createdAt?.toDate ?
post.createdAt.toDate().toLocaleDateString() : ""}
        </span>

        <Link
            to={` /post/${post.id}`}
            className="text-xs font-semibold text-blue-600
hover:text-blue-800"
        >
            Read More →
        </Link>
    </div>
</div>
</div>
    );
    })}
</div>
    )}
</div>
</div>
    );
}

```

src/pages/PostDetailPage.jsx (Public Detail View at /post/:id)

```

// src/pages/PostDetailPage.jsx
import { useState, useEffect } from "react";
import { useParams, Link } from "react-router-dom";
import { db } from "../lib/firebaseClient";
import { doc, getDoc } from "firebase/firestore";

export default function PostDetailPage() {
    const { id } = useParams();
    const [post, setPost] = useState(null);
    const [loading, setLoading] = useState(true);

    useEffect(() => {
        async function fetchPost() {
            try {
                const docRef = doc(db, "posts", id);
                const docSnap = await getDoc(docRef);

                if (docSnap.exists()) {
                    setPost({ id: docSnap.id, ...docSnap.data() });
                }
            }
        }
    });
}

```



```

    }
  } catch (error) {
    console.error("Error loading post:", error);
  } finally {
    setLoading(false);
  }
}

fetchPost();
}, [id]);

if (loading) {
  return (
    <div className="flex justify-center items-center min-h-[300px]">
      <p className="text-gray-500 animate-pulse font-medium">Loading post
content...</p>
    </div>
  );
}

if (!post) {
  return (
    <div className="text-center py-12 space-y-4 bg-white rounded-xl
border p-8 max-w-2xl mx-auto shadow-sm">
      <p className="text-lg font-semibold text-gray-800">Post Not
Found</p>
      <p className="text-sm text-gray-500">The article you are looking
for might have been deleted or moved.</p>
      <Link to="/" className="inline-block px-4 py-2 bg-blue-600 text-
white rounded-lg text-sm font-semibold hover:bg-blue-700 transition-
colors">
        ← Back to Blog Home
      </Link>
    </div>
  );
}

return (
  <article className="max-w-4xl mx-auto my-6 p-6 md:p-10 bg-white
rounded-2xl border border-gray-200 shadow-sm space-y-8 overflow-hidden">
    </* Navigation Header */>
    <div>
      <Link to="/" className="inline-flex items-center text-sm font-
semibold text-blue-600 hover:underline gap-1">
        ← Back to All Posts
      </Link>
    </div>

    </* Article Title & Metadata */>
    <div className="space-y-3 border-b border-gray-100 pb-6">
      <h1 className="text-3xl md:text-5xl font-extrabold text-gray-900
tracking-tight break-words leading-tight">
        {post.title}
      </h1>
    </div>
  </article>
);

```

```

        <div className="text-xs md:text-sm text-gray-500 flex flex-wrap
justify-between items-center gap-2 pt-2">
            <span className="font-medium">Author: <strong className="text-
gray-700">{post.authorEmail}</strong></span>
            {post.createdAt?.toDate && (
                <span>Published on
{post.createdAt.toDate().toLocaleDateString(undefined, { year: 'numeric',
month: 'long', day: 'numeric' })}</span>
            )}
        </div>
    </div>

    {/* Styled Article Content Body */}
    <div
        className="
            text-gray-800
            leading-relaxed
            space-y-4
            break-words
            overflow-hidden
            [&_img]:max-w-full
            [&_img]:h-auto
            [&_img]:rounded-xl
            [&_img]:my-6
            [&_img]:shadow-md
            [&_img]:mx-auto
            [&_h1]:text-3xl [&_h1]:font-bold [&_h1]:text-gray-900 [&_h1]:mt-6
            [&_h1]:mb-3
            [&_h2]:text-2xl [&_h2]:font-bold [&_h2]:text-gray-900 [&_h2]:mt-5
            [&_h2]:mb-2
            [&_h3]:text-xl [&_h3]:font-semibold [&_h3]:text-gray-900
            [&_h3]:mt-4 [&_h3]:mb-2
            [&_p]:mb-4 [&_p]:leading-relaxed
            [&_ul]:list-disc [&_ul]:pl-6 [&_ul]:mb-4
            [&_ol]:list-decimal [&_ol]:pl-6 [&_ol]:mb-4
            [&_blockquote]:border-l-4 [&_blockquote]:border-blue-500
            [&_blockquote]:pl-4 [&_blockquote]:italic [&_blockquote]:text-gray-600
            [&_blockquote]:my-4
            [&_a]:text-blue-600 [&_a]:underline
            "
        dangerouslySetInnerHTML={{ __html: post.content }}
    />
    </article>
);
}

```

src/pages/AdminPortalPage.jsx (Admin Dashboard at /admin)

```

import { useState, useEffect } from "react";
import { Link } from "react-router-dom";
import { db } from "../lib/firebaseClient";

```

```

import { collection, getDocs, query, orderBy, limit } from
"firebase/firestore";
import { StatCard, RecentPostsTable } from
"../components/DashboardComponents";

export default function AdminPortalPage({ user }) {
  const [stats, setStats] = useState({
    totalPosts: 0,
    myPosts: 0,
    postsWithImages: 0,
    lastPostDate: "N/A",
  });
  const [recentPosts, setRecentPosts] = useState([]);
  const [loading, setLoading] = useState(true);

  useEffect(() => {
    async function fetchAdminStats() {
      setLoading(true);
      try {
        // Fetch all posts to calculate accurate stats
        const postsRef = collection(db, "posts");
        const q = query(postsRef, orderBy("createdAt", "desc"));
        const snapshot = await getDocs(q);
        const postData = snapshot.docs.map((doc) => ({
          id: doc.id,
          ...doc.data(),
        }));

        // Calculate Real Data Statistics
        const total = postData.length;
        const myCount = postData.filter((p) => p.authorEmail ===
user?.email).length;

        // Count how many posts have at least one <img> tag in their HTML
content
        const withImages = postData.filter((p) => {
          if (!p.content) return false;
          const parser = new DOMParser();
          const doc = parser.parseFromString(p.content, "text/html");
          return doc.querySelector("img") !== null;
        }).length;

        // Determine latest published date string
        let latestDate = "No posts yet";
        if (postData.length > 0 && postData[0].createdAt?.toDate()) {
          latestDate =
postData[0].createdAt.toDate().toLocaleDateString();
        }

        setStats({
          totalPosts: total,
          myPosts: myCount,
          postsWithImages: withImages,
          lastPostDate: latestDate,

```

```

    });

    // Take top 5 latest posts for the preview table
    setRecentPosts(postsData.slice(0, 5));
  } catch (error) {
    console.error("Error loading admin stats:", error);
  } finally {
    setLoading(false);
  }
}

fetchAdminStats();
}, [user]);

return (
  <div className="space-y-6 pl-3">
    {/* Top Banner */}
    <div className="flex flex-col sm:flex-row sm:items-center sm:justify-between bg-white p-6 rounded-xl border border-gray-200 shadow-sm gap-4">
      <div>
        <h1 className="text-2xl font-bold text-gray-900 tracking-tight">Overview</h1>
      </div>

      <Link
        to="/admin/posts"
        className="px-4 py-2 bg-blue-600 text-white text-sm font-semibold rounded-lg hover:bg-blue-700 transition-colors shadow-sm text-center self-start sm:self-auto"
      >
        Manage All Posts
      </Link>
    </div>

    {/* Real Statistics Grid Cards */}
    <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-4">
      <StatCard
        title="Total Published Posts"
        value={loading ? "...": stats.totalPosts}
        color="text-blue-600"
      />
      <StatCard
        title="Authored by You"
        value={loading ? "...": stats.myPosts}
        color="text-green-600"
      />
      <StatCard
        title="Posts with Images"
        value={loading ? "...": stats.postsWithImages}
        color="text-purple-600"
      />
      <StatCard
        title="Latest Activity"

```

```

        value={loading ? "... " : stats.lastPostDate}
        color="text-amber-600"
        isText={true}
      />
    </div>

    { /* Live Recent Posts Table */ }
    <RecentPostsTable posts={recentPosts} loading={loading} />
  </div>
);
}

```

src/pages/PostsManager.jsx (Manage Posts at /admin/posts)

```

import { useState, useEffect } from "react";
import { db } from "../lib/firebaseClient";
import {
  collection,
  addDoc,
  getDocs,
  doc,
  updateDoc,
  deleteDoc,
  serverTimestamp,
  query,
  orderBy
} from "firebase/firestore";
import PostForm from "../components/PostForm";
import PostList from "../components/PostList";

export default function PostsManager({ user }) {
  const [posts, setPosts] = useState([]);
  const [loading, setLoading] = useState(true);

  // Navigation View Mode: "list" or "form"
  const [viewMode, setViewMode] = useState("list");
  const [editingPost, setEditingPost] = useState(null);

  const postsRef = collection(db, "posts");

  async function fetchPosts() {
    setLoading(true);
    try {
      const q = query(postsRef, orderBy("createdAt", "desc"));
      const querySnapshot = await getDocs(q);
      const fetchedPosts = querySnapshot.docs.map((docItem) => ({
        id: docItem.id,
        ...docItem.data(),
      }));
      setPosts(fetchedPosts);
    } catch (error) {

```

```
        console.error("Error fetching posts:", error);
    } finally {
        setLoading(false);
    }
}

useEffect(() => {
    fetchPosts();
}, []);

// Trigger form view for brand new post
function handleOpenCreate() {
    setEditingPost(null);
    setViewMode("form");
}

// Trigger form view for existing post
function handleOpenEdit(post) {
    setEditingPost(post);
    setViewMode("form");
}

// Handle Save (Create or Update)
async function handleSavePost(formData) {
    try {
        if (editingPost) {
            const postDoc = doc(db, "posts", editingPost.id);
            await updateDoc(postDoc, {
                title: formData.title,
                content: formData.content,
                updatedAt: serverTimestamp(),
            });
        } else {
            await addDoc(postsRef, {
                title: formData.title,
                content: formData.content,
                authorEmail: user?.email || "Admin",
                createdAt: serverTimestamp(),
            });
        }

        setViewMode("list"); // Return to table view
        setEditingPost(null);
        fetchPosts();
    } catch (error) {
        console.error("Error saving post:", error);
    }
}

// Handle Delete
async function handleDeletePost(id) {
    if (!window.confirm("Are you sure you want to delete this post?"))
        return;
}
```

```

    try {
      await deleteDoc(doc(db, "posts", id));
      fetchPosts();
    } catch (error) {
      console.error("Error deleting post:", error);
    }
  }

  return (
    <div className="space-y-8 max-w-5xl mx-auto">
      <div>
        <h1 className="text-2xl font-bold text-gray-900">Blog Posts
Management</h1>
        <p className="text-xs text-gray-500 mt-0.5">Create, update, and
manage public blog posts.</p>
      </div>

      {/* Conditionally renders either Table List View or Form View */}
      {viewMode === "list" ? (
        <PostList
          posts={posts}
          loading={loading}
          onCreateNew={handleOpenCreate}
          onEdit={handleOpenEdit}
          onDelete={handleDeletePost}
        />
      ) : (
        <PostForm
          editingPost={editingPost}
          onSubmit={handleSavePost}
          onCancel={() => setViewMode("list")}
        />
      )}
    </div>
  );
}

```

src/pages/LoginPage.jsx

```

import { useState } from "react";
import { Link, useNavigate } from "react-router-dom";
import { auth } from "../lib/firebaseClient";
import { signInWithEmailAndPassword } from "firebase/auth";

export default function LoginPage() {
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");
  const [loading, setLoading] = useState(false);
  const [error, setError] = useState(null);

  const navigate = useNavigate();

```

```

async function handleLogin(e) {
  e.preventDefault();
  setError(null);
  setLoading(true);

  try {
    await signInWithEmailAndPassword(auth, email, password);
    navigate("/admin"); // Directly navigates to Admin Portal!
  } catch (authError) {
    console.error("Firebase login error:", authError);
    setError(authError.message.replace("Firebase: ", ""));
  } finally {
    setLoading(false);
  }
}

return (
  <section className="flex-1 flex items-center justify-center bg-gray-50
px-4 py-12">
    <div className="w-full max-w-md bg-white rounded-2xl border border-
gray-200 p-8 shadow-sm">
      <h2 className="text-3xl font-bold text-center text-gray-800 mb-
1">Welcome Back</h2>
      <p className="text-center text-sm text-gray-500 mb-6">Login to
access AdminPortal</p>

      <form className="space-y-5" onSubmit={handleLogin}>
        <div>
          <label className="block text-sm font-medium text-gray-700 mb-
1">Email Address</label>
          <input
            type="email"
            placeholder="you@example.com"
            value={email}
            disabled={loading}
            onChange={(e) => setEmail(e.target.value)}
            required
            className="w-full rounded-lg border border-gray-300 px-4 py-
2.5 text-sm focus:outline-none focus:ring-2 focus:ring-blue-600
focus:border-blue-600"
          />
        </div>

        <div>
          <label className="block text-sm font-medium text-gray-700 mb-
1">Password</label>
          <input
            type="password"
            placeholder="....."
            value={password}
            disabled={loading}
            onChange={(e) => setPassword(e.target.value)}
            required

```



```

        className="w-full rounded-lg border border-gray-300 px-4 py-
2.5 text-sm focus:outline-none focus:ring-2 focus:ring-blue-600
focus:border-blue-600"
      />
    </div>

    {error && <p className="text-sm text-red-600 text-center">{error}
</p>}}

    <button
      type="submit"
      disabled={loading}
      className="w-full rounded-lg bg-blue-600 py-2.5 text-white
font-semibold hover:bg-blue-700 active:scale-[0.99] transition-all
disabled:bg-gray-400"
    >
      {loading ? "Signing in..." : "Sign In"}
    </button>
  </form>

  <p className="mt-6 text-center text-sm text-gray-600">
    Don't have an account?{" "}
    <Link to="/register" className="font-medium text-blue-600
hover:underline">
      Register
    </Link>
  </p>
</div>
</section>
);
}

```

src/pages/RegisterPage.jsx

```

import { useState } from "react";
import { Link, useNavigate } from "react-router-dom";
import { auth } from "../lib/firebaseClient";
import { createUserWithEmailAndPassword } from "firebase/auth";

export default function RegisterPage() {
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");
  const [confirmPassword, setConfirmPassword] = useState("");
  const [loading, setLoading] = useState(false);
  const [error, setError] = useState(null);

  const navigate = useNavigate();

  async function handleRegister(e) {
    e.preventDefault();
    setError(null);

```

```

    if (password !== confirmPassword) {
      setError("Passwords do not match");
      return;
    }

    if (password.length < 6) {
      setError("Password must be at least 6 characters");
      return;
    }

    setLoading(true);

    try {
      await createUserWithEmailAndPassword(auth, email, password);
      navigate("/admin"); // Directly navigates to Admin Portal!
    } catch (authError) {
      console.error("Firebase registration error:", authError);
      setError(authError.message.replace("Firebase: ", ""));
    } finally {
      setLoading(false);
    }
  }

  return (
    <section className="flex-1 flex items-center justify-center bg-gray-50 px-4 py-12">
      <div className="w-full max-w-md bg-white rounded-2xl border border-gray-200 p-8 shadow-sm">
        <h1 className="text-3xl font-bold mb-1 text-center text-gray-800">Register</h1>
        <p className="text-center text-sm text-gray-500 mb-6">Create a new Admin account</p>

        <form onSubmit={handleRegister} className="space-y-5">
          <div>
            <label className="block text-sm font-medium text-gray-700 mb-1">Email Address</label>
            <input
              type="email"
              value={email}
              disabled={loading}
              onChange={(e) => setEmail(e.target.value)}
              placeholder="you@example.com"
              required
              className="w-full rounded-lg border border-gray-300 px-4 py-2.5 text-sm focus:outline-none focus:ring-2 focus:ring-blue-600 focus:border-blue-600"
            />
          </div>

          <div>
            <label className="block text-sm font-medium text-gray-700 mb-1">Password</label>

```

```

        <input
          type="password"
          value={password}
          disabled={loading}
          onChange={(e) => setPassword(e.target.value)}
          placeholder="....."
          required
          className="w-full rounded-lg border border-gray-300 px-4 py-
2.5 text-sm focus:outline-none focus:ring-2 focus:ring-blue-600
focus:border-blue-600"
        />
      </div>

      <div>
        <label className="block text-sm font-medium text-gray-700 mb-
1">Confirm Password</label>
        <input
          type="password"
          value={confirmPassword}
          disabled={loading}
          onChange={(e) => setConfirmPassword(e.target.value)}
          placeholder="....."
          required
          className="w-full rounded-lg border border-gray-300 px-4 py-
2.5 text-sm focus:outline-none focus:ring-2 focus:ring-blue-600
focus:border-blue-600"
        />
      </div>

      {error && <p className="text-red-600 text-center text-sm">{error}</p>}}

      <button
        type="submit"
        disabled={loading}
        className="w-full bg-blue-600 text-white py-2.5 rounded-lg
font-semibold hover:bg-blue-700 active:scale-[0.99] transition-all
disabled:bg-gray-400"
      >
        {loading ? "Creating account..." : "Register"}
      </button>
    </form>

    <p className="mt-6 text-center text-sm text-gray-600">
      Already have an account?{" "}
      <Link to="/login" className="font-medium text-blue-600
hover:underline">
        Login
      </Link>
    </p>
  </div>
</section>

```

```
);  
}
```

Note: please follow previous lab to install tailwindcss and react-router

- Install Tailwind CSS `npm install tailwindcss @tailwindcss/vite`
- Git repository: https://github.com/rupp-ite/wct1_reactjs/tree/part3